Making a sign chart recipe

A *sign chart*, as we learned in class by working on this handout, is a labeled number line. Here's an example from the textbook, also. Here's a recipe for making them:

1. Figure out all the critical numbers of $f(x)$:
   - Find the derivative $f'(x)$ and figure out where $f'(x) = 0$ or does not exist.
   - For finding zeroes, it may be useful to review factoring and the Zero Product Property.
   - For finding places where $f'(x)$ does not exist, think about what could be *bad* x-values. Maybe are there things that would make you divide by zero? Or take the square root of a negative number? Or other weird domain issues?
   - Remember that the critical numbers are *inputs*.
2. Put the critical numbers on a number line.
   - We low-key don't care about any other numbers. In particular, we low-key don't care about endpoints, at least not for now.
   - The critical numbers chop up the number line into a bunch of segments between them.
3. Choose your favorite number from each of the segments and plug it into $f'(x)$.
   - We are making a number line about *slopes*, so we plug into $f'(x)$, not $f(x)$.
   - We low-key don't care what *number* comes out of $f'(x)$; we really only care about its *sign*: $+$ or $-$. (That's why this is called a *sign* chart!)
4. Label each segment with the *sign* of $f'(x)$ on that segment.
5. Translate the *signs* of $f'(x)$ into the *directions* – increasing or decreasing – of $f(x)$.

Finding extreme values recipe

I like the version of this recipe that we came up with in class even better than the fancy version that is in the textbook. Here's our in-class version:

1. Figure out all the critical numbers of $f(x)$.
   - I am once again emphasizing that the critical numbers are *inputs*.
2. Plug the critical numbers into the original function $f(x)$.
   - The outputs that result are called the *critical values*.
   - What would happen if you accidentally plugged the critical numbers into $f'(x)$?
3. The endpoint numbers are also inputs, so plug them into the original function $f(x)$ too.
4. Compare all these original $f(x)$ outputs and find the biggest and the smallest.

PS#10: Global optimization

For each of the functions below, which come from Activity 3.5.3:

1. Find the critical numbers by hand using algebra.
2. Draw a sign chart.
3. Find the absolute maximum and absolute minimum using the "finding extreme values recipe."
4. Finally, check your work by using Desmos to graph the function on the restricted domain.
   - Remember you can use the curly brackets to indicate a restricted domain.
   - Include a Desmos screenshot in your work for each of these six functions.

Here are the functions to do these four steps on:

(a) $h(x) = xe^{-x}$ on the interval $[0,3]$

(b) $p(t) = \sin(t) + \cos(t)$ on the interval $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

(c) $q(x) = \dfrac{x^2}{x-2}$ on the interval $[3,7]$

(d) $f(x) = 4 - e^{-(x-2)^2}$ with no domain restriction (so, no endpoints!)

(e) $h(x) = xe^{-ax}$ on the interval $\left[0, \frac{2}{a}\right]$, where $a > 0$ is some constant

(f) $f(x) = b - e^{-(x-a)^2}$ with no domain restriction, where a and b are some positive constants

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For part (e) and part (f), when you graph these in Desmos, you will probably be prompted to create sliders. This is a fun way to play around and see what different values of a and b do to the shape of the graph.